



Practice paper

Agentic AI: The building blocks of sophisticated AI business applications

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Abstract

Language models (LMs) like GPT-4's ability to predict word sequences have made tasks such as summarisation and translation significantly easier. They often struggle, however, with complex reasoning tasks that require deliberate, multi-step processes. To address these limitations, the concept of agentic artificial intelligence (agentic AI) is introduced, where LMs are organised into workflows that mimic human-like iterative reasoning. This paper explores the four pillars of agentic AI frameworks: tool use, reflection, planning and multi-agent collaboration (MAC). Tool use allows LMs to access external resources such as search engines to enhance accuracy; reflection enables self-correction and iterative feedback; planning ensures the structured breakdown of tasks; and MAC involves multiple LMs collaborating on specific subtasks. The paper then compares and contrasts various MAC frameworks based on the inclusion of tool use, reflection and planning abilities, as well as the flow and control of information between different agents. Finally, potential use cases for MAC frameworks are discussed.

Keywords: agentic AI; language models; multi-agent collaboration; complex problem solving

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Author's Biography

Chhavi Chawla is a data science professional with around nine years' experience in the analytics space. In MathCo, she leads delivery projects in generative AI across different domains of fast-moving consumer goods (FMCG), automotive and health.

Siddharth Chatterjee is a data science and machine learning (ML) graduate from UCL with more than nine years of experience in the analytics space. He leads the innovation team at MathCo in developing cutting-edge artificial intelligence products for their clients.

Sanketh Siddanna Gadadinni holds a Bachelor's degree in electronics and communication. With four years of experience, he specialises in building natural language processing (NLP) products and generative AI (GenAI) applications.

Pulkit Verma has over six years of experience in innovating products through artificial intelligence. He completed his Master's and Bachelor's degrees in the field of computer science engineering.



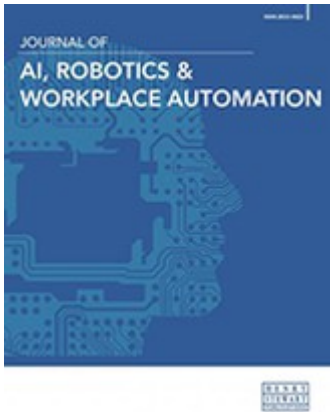
Sourav Banerjee is a Partner at MathCo, a company at the forefront of the generative AI landscape. With a focus on creating artificial intelligence (AI)-driven products tailored to customer needs, he is dedicated to developing a unique suite of offerings in GenAI. His expertise extends across several key areas of AI, including natural language processing, computer vision and forecasting.

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